

SAP-1

(SAP 1 is the simplest block design)

Simple As Possible (SAP 1) is a type of computer architecture that is used for academic purposes only. SAP-1 can only perform addition and subtraction not logical operations and these arithmetic operations are handled by the adder/subtractor unit. It has registers used for storing results and operands. The accumulator registers contains the result of the execution while the controller sequencer controls the operation of the computer. The output of the execution can be displayed at the Binary Display which has a 8-bit LEDs. The figure below is an architecture of SAP 1.

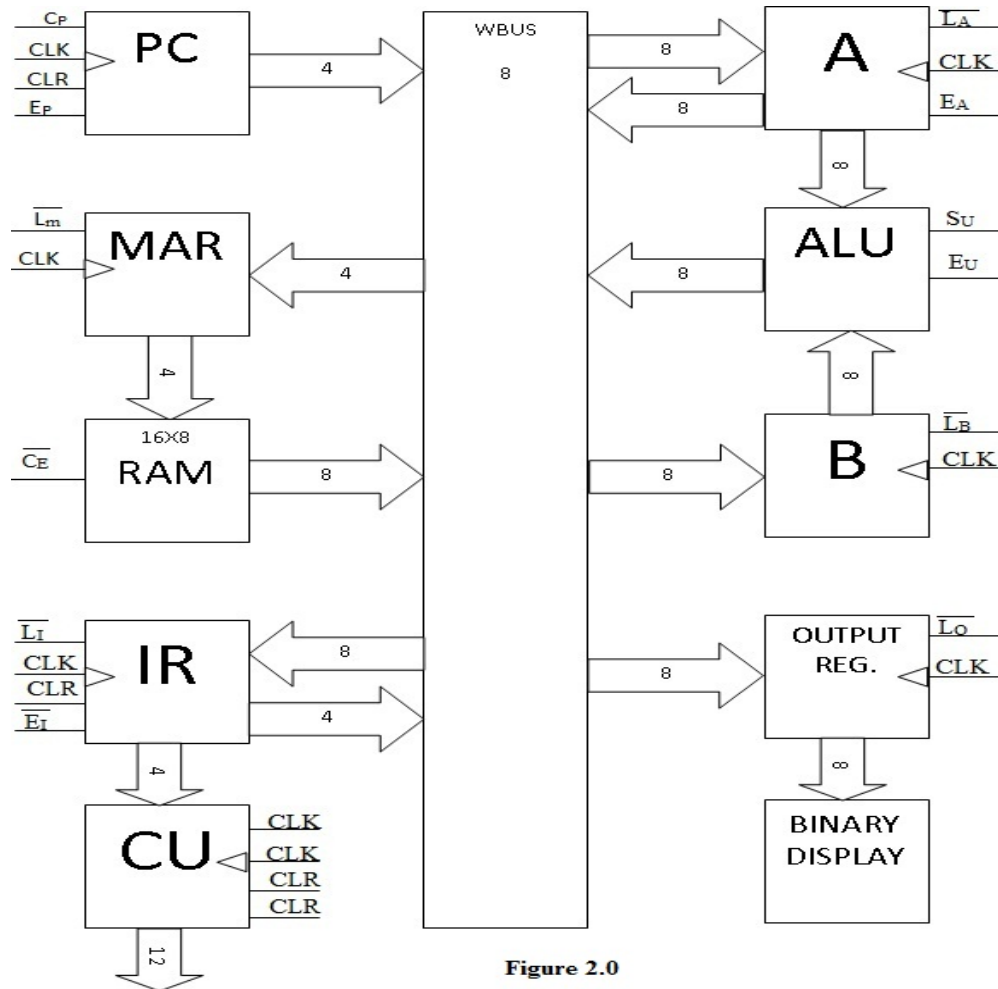


Figure 2.0

Figure 2: SAP - 1 Architecture

These are the SAP-1 architecture components:

(1) **Program Counter (PC):** It is a part of the control unit that counts from 0000 to 1111 i.e. from 0 to F in Hex form. Its job is to send address to the memory. The address of the next instruction to be fetched is executed.

(2) **Input and Memory Address Register (MAR)** includes the address and switch registers. These switch registers which are part of the input unit will allow us to send address bits to the RAM.

(3) **Static TTL RAM:** it allows us to store data in memory before a computer or program runs.

(4) **Instruction Register (IR):** It is a part of a control unit. The contents of the instruction register are split into two nibbles. The Upper Nibble is a 2 state output that goes directly to the Controller Sequencer which controls the operation of the computer. The 12 wires carrying the word are called Control Bus. The Lower Nibble is a 3 state output that is directed onto the WBus.

(5) **Accumulator:** It is a buffer register that stores intermediate result of a computer operation. It has 2 outputs, the two-state output goes directly to the adder/subtractor and the three-state output goes to the WBus.

(6) **ALU:** It uses 2's complement Adder/Subtractor. When Su is low and the sum out of the adder-subtractor is $S=A+B$. When Su is high, the difference appears $A=A-B$, and when Eu is high, the contents appear on the WBus.

(7) **B-Register:** It is a buffer register used in arithmetic operations.

(8) **The output register:** The accumulator contains the result of the executions and its content can be loaded into the Output Register. It is an interface to the outside system of the computer.

(9) **Binary display:** It is a row of 8-bit LEDs where the result can be displayed.

The SAP-1 architecture has only 16 x 8 RAM. Aside from this, the RAM is a static Transistor-Transistor-Logic (TTL) The bus of SAP-1 is 8-bit wide which is similar to Intel 8008 microprocessor which can access 8-bits of data in a single instruction. Every computer has an instruction cycle or the fetch/execute cycle. The CPU gets an instruction from the memory (fetch) and then perform necessary action to the instruction (execute). After the instruction has been executed, it can either halt or fetch the next instruction. The SAP-1 is a synchronous circuit and executes one instruction at a time. Even early microcomputers follows the same fetch/execute instruction cycle. SAP-1 has five instructions which are all stored in the memory.

Mnemonics	Opcode	Syntax	Operation
LDA	0000	LDA x	[x] $\rightarrow$ ACC
ADD	0001	ADD x	[x] $\rightarrow$ B ACC $\leftarrow$ [ACC] + [B]
SUB	0010	SUB x	[x] $\rightarrow$ B ACC $\leftarrow$ [ACC] - [B]
OUT	1110	OUT	[ACC] $\rightarrow$ 0
HLT	1111	HLT	Disable

Table 1: Instruction Set of SAP 1 Architecture



The Mnemonics codes **LDA** (load into accumulator), **ADD** (add memory to accumulator), **SUB** (subtract memory from accumulator), **OUT** (display data) and **HLT** (stop execution) are all assembly language. On the other hand, *opcode* is the equivalent machine language of each mnemonic. Opcode identifies the instruction to be executed and each instruction has its unique opcode that the computer can handle. SAP-1 instruction cycle is broken into two cycles, the first is the fetch cycle which makes use of the PC, MAR, IR and RAM while the other one is the execution cycle. Fetch and execute cycles have three timing states each, T1, T2, and T3 for fetch cycle and T4, T5, and T6 for execute cycle. Each component of the SAP-1 has its own control lines which need to be activated and they can be either active high or active low. Activating the control lines enable each component to communicate with a particular device and the control inputs are handled by the control unit.

SAP2

SAP2 is next evolution towards modern computer because it can perform addition and subtraction and also the new instruction called "jump". (SAP1 is the primitive form of computing machine.) This new instruction forces the computer to repeat or skip parts of the program.

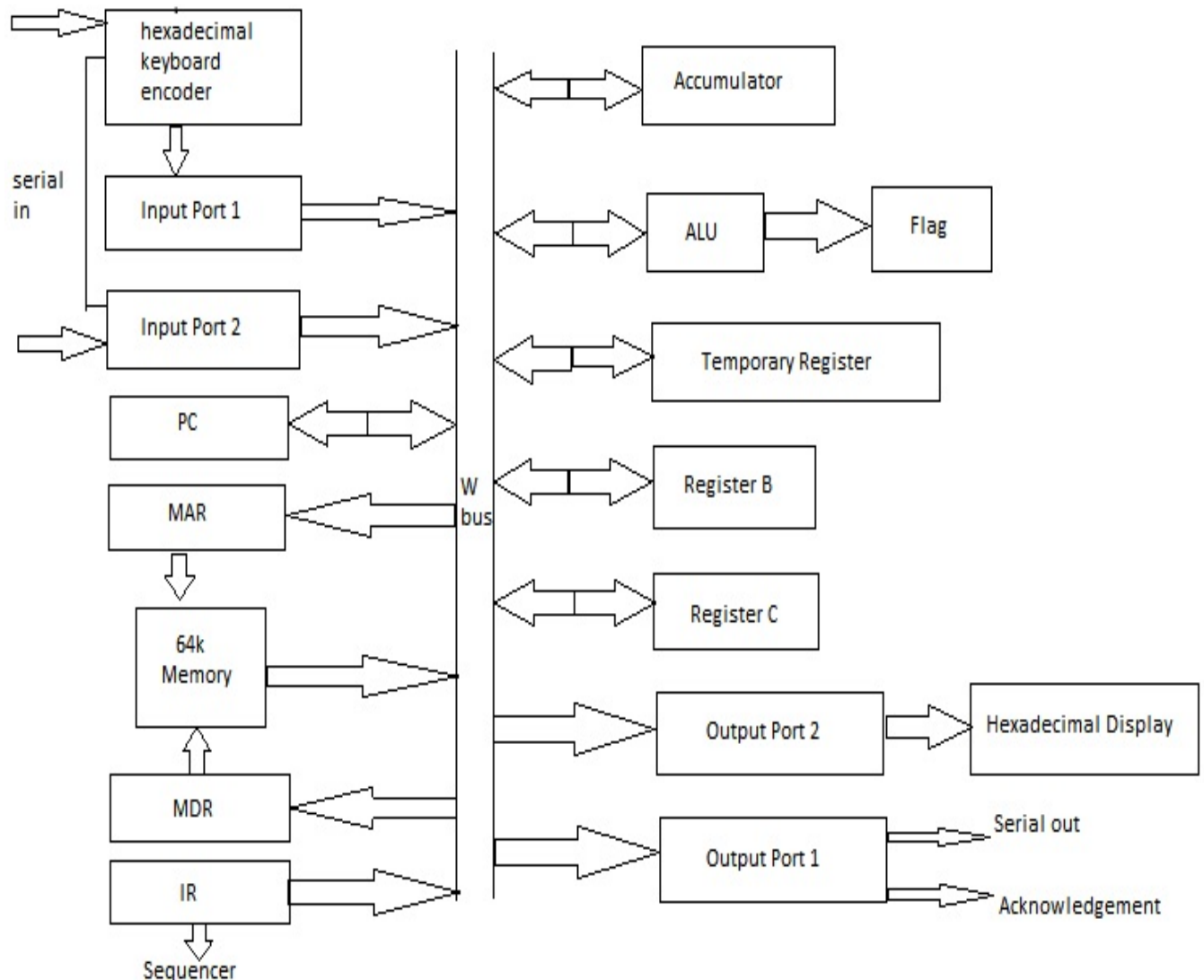


Figure 1: SAP - 1 Architecture

These are the SAP-2 architecture components:

(1) Bidirectional registers: to reduce the wiring capacitance of SAP2 we run only one set of wire between each registers and bus. In bidirectional registers data can flow in both ways.

(2) Input port: SAP2 has two input ports, number 1 and number 2. A hexadecimal keyboard encoder is connected to port 1. It allows us to enter hexadecimal instructions and data through port 1.

(3) Program counter: it has 16 bit so it can count from 0000H to FFFFH (0 to 65536). It sends address to MAR.



(4)MAR and memory: MAR receives 16-bit address from the program counter. The two-state MAR output then addresses the desired memory location. The memory has a 2k ROM with address of 0000H to 07FFH. ROMM contain a program called a 'monitor' that initialize the computer. It has 62k RAM i.e. (0800H to FFFFH)

(5)Memory Data Register(MDR): it is an 8-bit buffer register. Its output sets up the RAM. The memory data register receives data from the bus before a write operation and it sends data to the bus after a read operation.

(6)Instruction Register(IR): It handles 8-bit opcode, so it can accommodate 256 instruction. SAP2 has only 42 instruction. These instructions are similar to that of 8085.

(7)Controller sequencer: this produce the control word or micro instruction that coordinate and direct the rest of the computer. Control word or micro instruction determines how the register react to the next positive clock cycle.

(8)Accumulator: Here the two state output goes to ALU, and three state output goes to W-bus. Words appear in bus when E_A is active.

(9)Flag and ALU: ALU have 4 or more control bits that determine the arithmetic or logic operation performed on words A and B. A flag is a flip-flop that keep track of a changing condition during a computer run. It has two flags i.e. sign flag and zero flag.

(10)TMP,B and C register: instead of using the B register to hold the data being added or subtracted from the accumulator, a temporary (TMP) register is used. B register can be used freely. It also consists of C register. This C register can also be used freely for storing data and performing ALU operations.

(11)Output ports: SAP-2 has two output ports numbered 3 and 4. The content of accumulator can be loaded into port3. It also consists of serial port output.

Instruction set of SAP2: the instruction set of SAP 2 consists of MOV, MVI, INR, DCR, CMA, ADD, ANA, ANI, HLT, IN, OUT, JM, JZ, JNZ, LDA, STA, SUB, XRA, XRI, ORA, ORI, RAL, RAR, RET,NOP.

1) Draw the block diagram of basic microprocessor and explain it. Which block design is simple and explain it?

Answer: (Both SAP-1 and SAP-2 are referred as basic microprocessor)